

“DOUBLE-CONVERSION UPS WILL CONTINUE TO DO WELL IN INDIA”

Venkat Rao is the country manager for medium and large UPS systems at Emerson Network Power (India). In this position, he is responsible for defining product roadmap, product lifecycle management, new product introduction, business growth with profitability, cost and price management, identifying markets for business growth, product promotion, resource management and competency building, and competition analysis.

Rao speaks to Uma Bansal, deputy editor at EFY, about the trends in UPS systems. Excerpts:

■ VENKAT RAO

COUNTRY MANAGER FOR MEDIUM AND LARGE UPS SYSTEMS AT EMERSON NETWORK POWER (INDIA)

At what stage should the decision to use a UPS system be taken up?

UPS systems are required for critical loads that cause downtime. The decision for a UPS can be taken at the system designing stage, system implementation stage, or when the load which causes disruptions in business processes is identified and magnanimity of revenue loss from downtime of such load is identified.

One should know whether uninterruptible power is essential for one's application. In some cases, power interruption does not make any difference to operation and the cost of downtime is negligible.

Cost of downtime and operating expenditure decide the type of power supply to be deployed. There are three types of needs: uninterruptible power without conditioning, uninterruptible power with conditioning and only conditioned power. Some equipment manufacturers require conditioned power.

What factors to consider when selecting a UPS?

The kVA/kW rating of the load, the backup time required and the uptime desired. Most importantly, the load, and its kW rating and power factor. Attention

needs to be paid to load characteristics like inrush current and crest factor, particularly for industrial loads. Load demand in terms of specifications and UPS deliverables should match. The backup time requirement has something to do with the bill of materials. The choice of topology and configuration depends on criticality/uptime requirement of the application.

What UPS technologies are available? Which applications do they suit the most?

Broadly, UPS systems can be categorised into static and dynamic types.

There are three standard technologies available in static UPS: Offline, line-interactive and online. Offline UPS systems are the least expensive of the three. These systems switch from the utility power to battery power only when the utility power fails. Generally available below 3 kVA, they are suitable for users looking for simple protection.

Line-interactive systems are similar to offline systems, except that these provide constant monitoring of the integrity of utility power and correct voltage when the input voltage fluctuates within a certain band. This technology is available in the sub-20kVA range and gives slightly greater protection than offline systems.

Online UPS systems, also known as double-conversion systems, are expensive owing to their complex design and functionalities. These units are available in the 3kVA range. Here all the power components are continuously on line, hence the name. As the incoming power is converted twice, during the rectification stage, input disturbances get eliminated and the load experiences clean and regulated voltage generated by the inverter. These UPS systems are useful for mission-critical business applications in all verticals.

Do factories require a special UPS design?

As organisations become more technology-dependent,

